

Pantelis Kotsas

Athens, Greece | pandeliskotsas@gmail.com | [LinkedIn](#) | [GitHub](#)

Profile

Computer Science & Telecommunications student at NKUA with hands-on experience in systems programming, kernel development, backend web development, and cloud-native architectures. Strong foundation in OS internals, algorithms, and data structures, with practical project experience spanning C kernel extensions, POSIX concurrency, disk-based storage engines, Python web backends, and Azure serverless platforms.

Education

B.Sc. in Computer Science and Telecommunications *National and Kapodistrian University of Athens (NKUA) · Athens, Greece · 2023 – Present*

Notable grades: Artificial Intelligence I (9/10) Algorithms & Data Structures (8/10) New Venture Creation (9.5/10)

Projects

HITL Gateway – AI Agent Governance Platform *Techbiz Hackathon, Microsoft Challenge · 2026*

Full-stack AI governance middleware on Azure Durable Functions with zero-compute wait states, multi-tenant isolation, and a full audit trail (filterable timeline + CSV export). Built a multi-stage SLA escalation pipeline (Telegram, Twilio voice pager, Resend), a real-time React dashboard with live orchestration visualization, and layered enterprise security (HMAC-SHA256, SSRF prevention, rate limiting, replay prevention) – 91 passing tests.

MLFQ Scheduler in xv6-riscv *Operating Systems Course, NKUA · 2025*

Extended the xv6-riscv teaching OS kernel with a custom multi-level feedback queue scheduler. Implemented 4-level priority-based preemption with timeslice demotion, an anti-starvation boost mechanism, and a `getpinfo()` system call end-to-end (kernel → syscall dispatch → user-space `ps` utility).

Disk-Based B+ Tree File Engine *Databases Course, NKUA · 2025*

Implemented a B+ tree in C over a 512-byte block file library supporting insert, split, and search. Built correct node splitting, tree rebalancing, and leaf-chain traversal; inserted and searched 400+ records while diagnosing and fixing critical memory corruption in block size constraints.

IPC Multi-Process Messaging System *Operating Systems Course, NKUA · 2025*

Built a multi-process messaging system in C using POSIX shared memory and named semaphores. Supported 10 concurrent dialogs with exactly-once delivery, mutual exclusion via `sem_wait/post`, non-blocking I/O polling via `select()`, and graceful shutdown with automatic resource cleanup.

HelpMeAnytime – Apps4Athens Hackathon 2.0 *Hackathon · Athens · 6th / 60 teams · 2025*

Architected a full civic platform (Next.js + TypeScript + PostgreSQL) connecting Athens residents with local artisans, volunteers, and neighborhood crowdfunding. Coordinated the development team under hackathon time pressure and pitched the business model to the jury.

Notable – Note-Taking App Startup Simulation *New Venture Creation Course, NKUA · 2025*

Led a simulated startup as CEO and CTO. Directed product strategy, market research, and business model canvas; gathered feedback from 50+ users via surveys and delivered a full business plan and elevator pitch to faculty.

Technical Skills

Programming Languages	C/C++, Python, TypeScript/JavaScript, RISC-V Assembly, SQL, HTML/CSS, MATLAB
Systems	xv6-riscv kernel, POSIX Shared Memory & Semaphores, Linux
Backend & Web	FastAPI, Next.js, React, Python (Bottle), Firebase, PostgreSQL, Tailwind CSS
Cloud & DevOps	Azure Durable Functions, Azure Event Grid, Twilio, Server-Sent Events, Docker
Tools	Git/GitHub, MySQL Workbench
Languages	Greek (Native), English (C2 – Cambridge CPE)

Certifications & Distinctions

Hack 'n Learn – Hellenic Cyber Security Team (Web & Binary Exploitation)	May 2024
Big Data Analytics and Web Development with AI – IEEE University of Piraeus	May 2024
Digital Marketing Fundamentals – IIDE	Jan 2024
Cambridge Proficiency in English (CPE) – C2 Level	2022
7th International Olympiad of Hellenism – Romania	Sep 2022